manned aircraft. These UCAVs are intended to have full autonomy. Here, the main human interaction with the aircraft is not piloting it but authorising firing of weapons.

Helicopter pilots. In some remote areas, power-line inspection is being carried out from helicopters with human pilots, but this can be replaced with AI-based autonomous drones.

How close is AI to replacing the human pilot

Robots replacing human pilots is not good news for the latter. In principle, an autonomous combat aircraft capable of carrying out air-to-air and ground-attack missions is technically feasible. And these autonomous aircrafts are already available in limited areas.

Like any other industry, the aviation industry will embrace cost-cutting using robots. With advanced AI technology, more and more aircraft controls are being entrusted to AI systems. Robots and cognitive machines are already more advanced than humans at most jobs.

But replacing human pilots by AI is a complex mission, and would involve technical, legal and political aspects. Auto-pilot feature to an extent can fly a plane but it generally involves some human interaction. Complete automation without human intervention is not an easy task. Moreover, to get an auto-pilot certified by a regulatory body is a massive undertaking.

Autonomous commercial aircraft would also require updating entire ATC systems. If an ATC needs to reroute the plane, or issue a change in course, currently it is handled via person-to-person communication. Some kind of command to the plane via data link would be needed to do it via an automated system. Thus, automation will be needed both in aircraft and the ATC system.

A UCAV has some intelligence along with the capability of flying to a pre-determined location. Dropping some mines and then returning to a base location might be militarily cost-effective. But we cannot compare auto-pilot features in commercial planes with UCAVs, because UCAVs have limited features and control functions.

There are many risks involved with pilot-less commercial planes. Therefore experts feel that fully-functional commercial planes without human pilots may not be feasible in the next decade or so.

Drawbacks of AI in aviation

While AI is promising for the future of the aviation industry, it has some pitfalls as well. First, AI is quite expensive. All airlines might not be able to afford or invest in such new and expensive technology.

Second, it will take some time for implementation by the worldwide aviation industry.

Recently, Boeing was under intense scrutiny after its 737 Max Jet was involved in two deadly crashes. In March, Ethiopian Airlines Flight 302, a Boeing 737 series crashed minutes after it took off, and all 157 people onboard died. The first such crash took place in October 2018 in Indonesia. Lion Air Flight 610 crashed just minutes after taking off from Jakarta, killing 189 people.

MCAS software installed in jets was the key reason behind the fatal accidents. These incidents are an eye-opener for the use of AI technology in the aviation sector.

Commercial aircrafts are complex systems, and pilot-less planes put into service is not an easy job. Even if such systems exist, only a few people are willing to fly in fully-autonomous aircraft. A survey done in 2018 has shown that just under thirty per cent of the US flyers are willing to fly on autonomous aircraft.

Conclusion

We are going to witness a huge change in the world of aviation through AI technology. Though AI is still at a nascent stage, we have already seen a number of changes in the aviation industry. Applications like crew management, flight management, ticketing, maintenance and passenger identification are all centred around improving customer experience.

Advances in AI like auto-pilot features are reshaping the future of the airline industry. And yet the notion that AI would fully replace human the pilot is still a distant dream. But yes, AI is going to improve overall airline operations and businesses in a big way! EFY